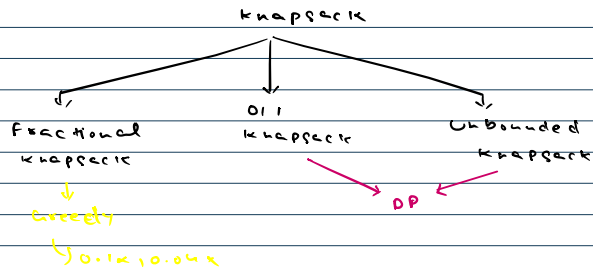


Dynamic Programming

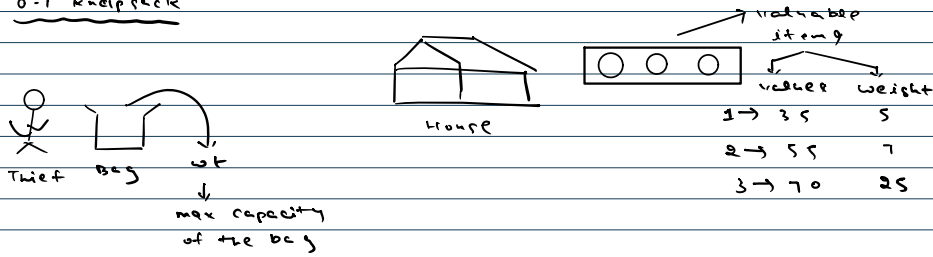
Enhanced Recursion
Memoised Recursion

overlapping subproblem
Optimal substructure

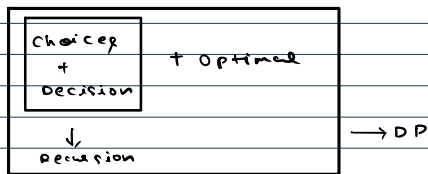
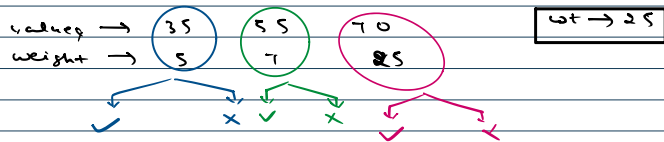
Look up
table



0-1 knapsack



profit maximum → optimal path
max value of the bag (sack)



wt [] : 1 3 4 5
val [] : 1 4 5 7
w : 7

→ Input

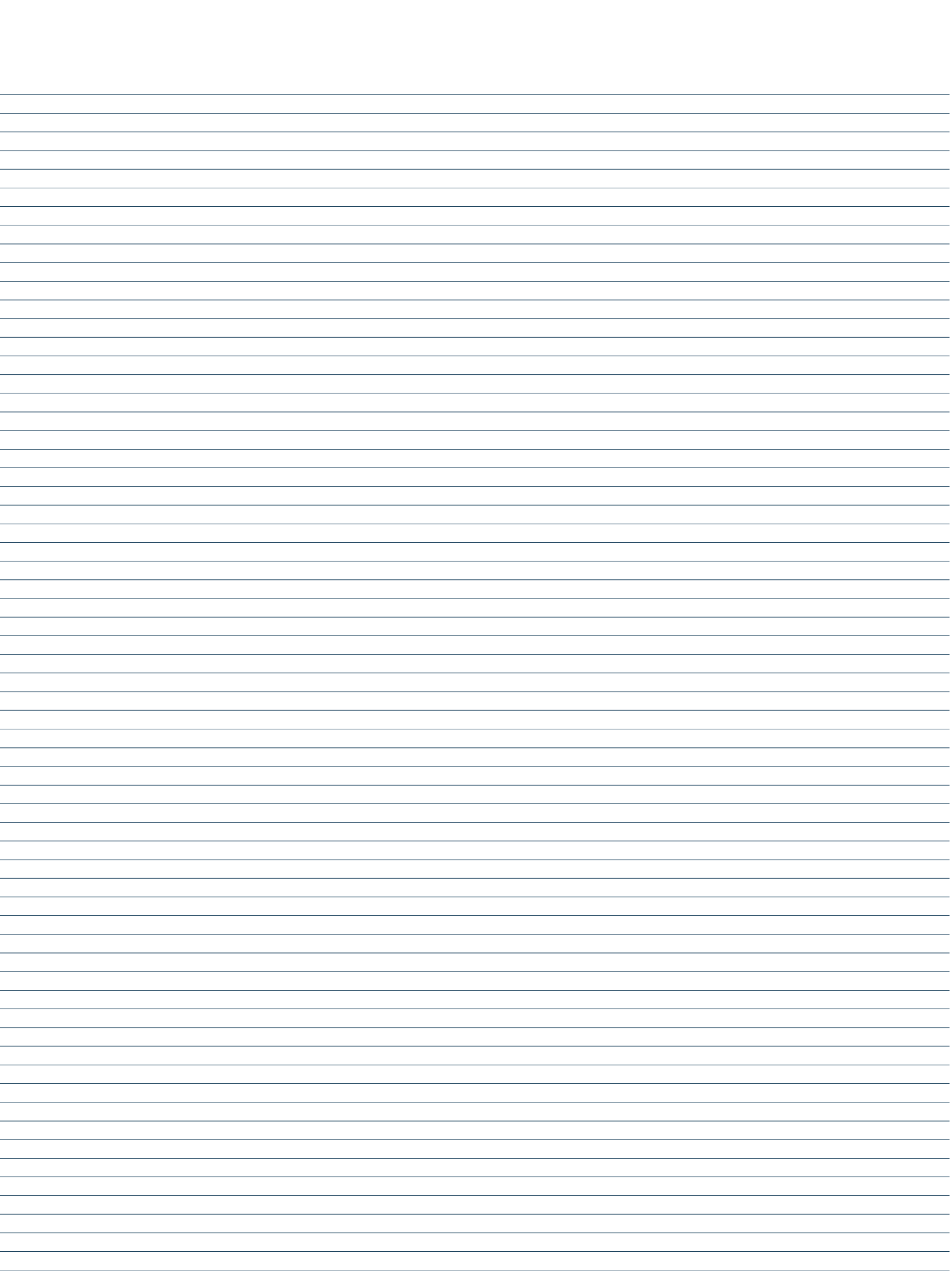
O/P :- max profit

Recursion → choices + decision → optimal } DP

Recursion → func(ip) → func(smaller ip)

Choice Diagram





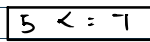
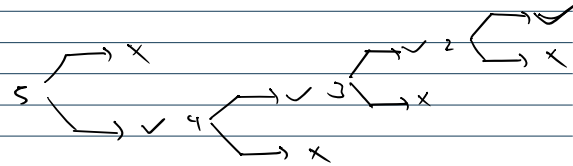
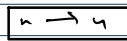
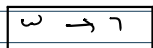


Base catch $\rightarrow$ think of smallest valid ip.

→ array size = 20

$$k_z = 0$$

Bag weight = 0

$$\omega = 0$$

$$\omega = 1, 3, 4, 5$$

val $\rightarrow 1, 4, 5, 7$

$$w = 7$$

7

x

1 2 3 4 5

$F \Rightarrow 1, 3, 4$

2-3, 4, 5

 $\rightarrow$

3

5-4-3

make matrix for those variables which are changing:

Input $\rightarrow$ n



n + 1

$$dp \in \omega^{\omega}$$

Look up Table

